

CityMind

AI-Powered Infrastructure Digital Twin Structural Inspection Report

Report ID:	TWIN-BRIDGE-20260228
Date:	February 28, 2026
Structure:	N/A
Health Index:	37.5/100 (Grade: F)
Status:	Critical

Executive Summary

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Defect Analysis

ID	Type	Severity	Confidence	Method	Code Reference
DEF-004	settlement	8.3/10	95%	yolov8	
DEF-017	settlement	9.1/10	94%	rt-detr	
DEF-018	corrosion	9.1/10	83%	yolov8	

Code Compliance & Risk Assessment

Compliance Status:	Non-Compliant
Total Violations:	16
Risk Score:	100/100
Risk Level:	High

AMD Technology Stack

AMD Instinct MI300X	Model training (ROCm 7.2)
AMD Ryzen AI NPU (XDNA)	Edge inference for perception models
Vitis AI Quantizer	INT8 model optimization
ONNX Runtime + Vitis AI EP	NPU deployment runtime
AMD GAIA Framework	Multi-agent orchestration
Lemonade (TurnkeyML)	LLM optimization pipeline
Ryzen AI Software v1.7	Local AI runtime (16K context)
AMD CVML Library	Depth estimation pipeline
Ryzen AI Max+ iGPU	3D reconstruction acceleration
Genesis Simulation Engine	Digital twin physics
Vulkan Backend	GPU rendering optimization
PyTorch (ROCm) + TunableOp	Training kernel optimization

Disclaimer: This report is generated by CityMind AI and is intended as a preliminary screening tool. All findings should be verified by a licensed Professional Engineer (PE). CityMind does not replace professional engineering judgment.